Zurich ServiceNow AI Platform Capabilities

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CMDB Compliance

CMDB Compliance is a tool set that enables administrators to certify CMDB data for correctness and fix any discrepancies found in the data.

Note: For compliance in the context of internal business goals and objectives, and external legislation and regulations, see Exploring Policy and Compliance Management.

Certification options

CMDB Compliance offers these certification options to suit the size and requirements of your organization:

Option	Description
Desired State	Automatically compares the actual attributes and relationships of specific ServiceNow records against the desired states for those records. For example, an audit can detect a Linux database server with insufficient RAM or whose Depends on relationships with another CI is incorrect. The system then publishes any discrepancies found and automatically assigns follow-on tasks to qualified users to bring that server into compliance.
Architecture Compliance	Automatically compares the actual attributes of specific CIs, such as CPU count, RAM, or disk size against the expected attributes for those CIs. The system publishes any discrepancies found and automatically assigns remediation tasks to qualified users.

Compliance Templates and Audits

The Templates and Audits modules on the top level of the Compliance menu enable a certification_admin user to create, edit, and delete all template and audit types.

You can use Compliance Templates and Audits to evaluate records for any table in the ServiceNow system, not just those tables extending the Configuration Item [`cmdb_ci`] table. Compliance audits certify record attributes only. Compliance templates can be used in Control Test Definitions in Governance Risk and Compliance.

Compliance Activation

Compliance functionality is provided by the Certification Core (com.snc.certification_core) plugin which contains shared functionality required for certification audits.

The Certification Core (com.snc.certification_core) plugin consists of the following plugins, and is activated by default.

- Activated by default: Desired State Certification (com.snc.certification_desired_state)
- Activate: Architecture Compliance (com.snc.architecture_compliance), which automatically activates the Version Management (com.snc.version) plugin that manages certification filter and template versions.
- Activate: Data Certification (com.snc.certification_v2), which automatically activates the Version Management (com.snc.version) plugin that manages certification filter versions.

Installed with Compliance

These components are installed with the Certification Core plugin.

Demo data is included with the Desired State and Architecture Compliance plugins.

The Certification Core plugin adds or modifies these tables.

Compliance Certification Core tables

Name	Description
Audit [cert_audit]	Contains all the data required to run an audit, including the users assigned to follow-on tasks and the run schedule.
Audit Result [cert_audit_result]	Contains the results of specific, certification audits.
Follow On Task [cert_follow_on_task]	Contains the tasks that were generated from an audit discrepancy.
Certification Template [cert_template]	Contains the definition of the desired state of the record. The template includes a filter that identifies the records to evaluate and the expected attributes and relationship values. Contains the records to certify, the expected attributes, and the expected relationship values.
Certification Condition [cert_cond]	Base table that defines the desired attribute or relationship conditions used in templates.
Certification Attribute Condition [cert_attr_cond]	Contains the conditions that define the desired CI attribute values. This table extends the Certification Condition [cert_cond] base table.
Certification CI Relationship Condition [cert_ci_rel_cond]	Contains the CI to CI relationship conditions. This table extends the

Name	Description
	Certification Condition [cert_cond] base table.
Certification User Relationship Condition [cert_user_rel_cond]	Contains the CI to user relationship conditions. This table extends the Certification Condition [cert_cond] base table.
Certification Group Relationship Condition [cert_group_rel_cond]	Contains the CI to group relationship conditions. This table extends the Certification Condition [cert_cond] base table.
Certification Related List Condition [cert_related_list_cond]	Contains the related list conditions. This table extends the Certification Condition [cert_cond] base table.
Certification Filter [cert_filter]	Contains a certification filter, including the table that contains the records to audit and the filter conditions.

User roles

The certification role is automatically assigned to all users with the itil role when the Certification Core plugin is activated or when compliance applications are upgraded. Certification core installs two business rules, both called **Add Certification Role To Manager**, that perform similar tasks on different tables. One rule checks for a manager specified on the User [sys_user] table, and the other checks for the certification role on the User Role [sys_user_has_role] table. When both a manager and the certification role are specified for a user, the system automatically grants the certification role to the manager. This functionality ensures that a certification task can be escalated successfully to the next level. The system grants this automatic role to the user's immediate manager only and not to others up the management chain.

> **Note:** When a manager has only the certification role and no other role, the manager is considered a Requester and is not counted as a subscribed user (Fulfiller).

Compliance Certification Core user roles

Name	Contains roles	Description
certification	none	Can read and update certification tasks to resolve discrepancies.
certification_filter_admin	none	Can create, read, and update certification filters.
certification_admin	certification, certification_filter_admin	Can manage the entire certification process. These users can create, edit, and delete all certification records.

UI policies

Compliance Certification Core UI policies

Name	Table	Description
Make table read only	Audit [cert_audit]	Sets the table field derived from the selected filter to read-only.
Hide Audit Type	Audit [cert_audit]	Hides the Audit type field.
Hide next scheduled run	Audit [cert_audit]	Hides the Next scheduled run date when an audit is inactive or on-demand.
Show task fields when create tasks is set to true	Audit [cert_audit]	Displays all fields related to creating tasks when the user

Name	Table	Description
		selects the Create tasks check box.
Make name mandatory	Audit [cert_audit]	Makes Name a mandatory field.
Prevent editing of Last run date	Audit [cert_audit]	Makes Last run date field read-only.
Show User field	Audit [cert_audit]	Shows or hides fields based on the Assignment type selected. The system shows the User field when you select the following assignment types: • User Field if the Assign to empty option is Create Assigned Task. • Specific User
Show Assign to fields	Audit [cert_audit]	Shows or hides fields based on the Assignment type selected. The system shows the Assign to field when the assignment type is User Field.
Show Assignment Fields	Audit [cert_audit]	Shows or hides fields based on the Assignment type selected. The system shows the Assign to empty field when you select either of the

Name	Table	Description
		following assignment types: • User Field • Group Field
Show Group field	Audit [cert_audit]	Shows or hides fields based on the Assignment type selected. The system shows the Group field when you select either of the following assignment types: • Specific Group • Group Field if the Assign to empty option is Create Assigned Task.
Hide "run" associated fields when active is set to false	Audit [cert_audit]	Hides these scheduling fields when the audit is inactive: • Run • Day • Time • Last scheduled run
Show script window on Scripted Audit	Audit [cert_audit]	Displays the Run this script field when the audit type is Scripted.

Name	Table	Description
Make table read only	Certification Condition [cert_cond]	Sets the table field derived from the selected filter to read-only.

Script includes

Compliance Certification Core script includes

Name	Description
DesiredStateUtil	Utility functions for desired state, used to clone a template for Insert functionality.
CMDBRElationshipAjax	Tool to get all relationships for a given table.
RelationshipQueryParseAjax	Parses condition filters. This script include is the internal code used in generating the compliance conditions.
CertificationUtils	Utility functions for certification that find Next run time value, and so on.
CertTaskEscalationTimerPercentag e	Utility method for setting escalation timer durations.
ConditionUtilsAjax	AJAX utilities for parsing queries into a human-readable format.
DeleteInactiveVersionsAjax	AJAX server-side script to delete all inactive versions of a record.

Client scripts

Compliance Certification Core client scripts

Name	Table	Description
Make audit type read only if not new	Certification Template [cert_template]	Sets the correct audit type for new records, and if the record is not new, sets the Audit type field to read only.
Update table name (filter)	Audit [cert_audit]	Updates the table Name field when the filter is updated.
Update table name	Audit [cert_audit]	Updates the table Name field when the template is updated.
Set table name on new	Audit [cert_audit]	Returns the table name from the template or filter.
Update table name	Certification Template [cert_template]	Updates the table Name field when a new filter is chosen and checks all existing conditions to see if they work for the new table.
Show conditions when table is set	Certification Template [cert_template]	Shows and hides conditions appropriately when the table is set.
Reset filter when audit type changes	Certification Template [cert_template]	Clears the filter and updates the lists shown when the audit type is changed.

Business rules

Compliance Certification Core business rules

Name	Table	Description
Clone condition	Certification Condition [cert_cond]	Part of certification versioning. This business rule retains the original ID when a condition is changed.
Copy audit type from audit	Audit Result [cert_audit_result]	Ensures that all audit results have the same audit type as the audit that generated them.
Copy values from template	Audit [cert_audit]	When a user selects a template, and updates the table, filter, and audit type from the template.
Delete condition	Certification Condition [cert_cond]	Part of certification versioning that deletes a condition.
Prevent deletion of audit with results	Audit [cert_audit]	Prevents deletion of an audit containing results.
Prevent delete of Filter with Template	Certification Filter [cert_filter]	Prevents deletion of a filter still linked to a template or audit.
Prevent deletion of result with task	Audit Result [cert_audit_result]	Prevents deletion of an audit result with an attached task.
Prevent delete of Template with Audit	Certification Template [cert_template]	Prevents deletion of a template still being used by an audit.

Name	Table	Description
Update conditions' tables	Certification Template [cert_template]	When storing template conditions, properly run all workflows and update the condition fields to contain the display version of the conditions.
Update filter version	Certification Filter [cert_filter]	Creates a version when the filter changes in any meaningful way.
Update next run time	Audit [cert_audit]	Updates the time in the Next scheduled run field when an audit is modified.
Update next run time during execution	Audit [cert_audit]	When the audit runs, update the Next scheduled run field to the next time the audit is scheduled to run.
Update table	Certification Template [cert_template]	Update the stored table to the table of the filter.
Update template version	Certification Template [cert_template]	Creates a version when the template changes in any meaningful way.

Compliance Overview module

The Compliance Overview module is a type of a homepage.

About this task

The Compliance Overview module summarizes:

- Current audit states
- Outstanding certification tasks
- Compliance discrepancies
- Upcoming audits
- General state of compliance audits for Data Certification, Desired State, Architectural Compliance, and Scripted audits

Only users with certain roles can access the Overview module. The different levels of access are:

Access levels per role

Role	Access
certification	View (view overview page and refresh reports)
certification_admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) View, customize
admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) • Edit (can edit gauges)

Procedure

1. Navigate to **All > Compliance > Overview**.

2. Select elements within the reports to obtain more information.
   For example, select a bar in the Compliance Discrepancies chart to open a list of audit results filtered by the respective attributes.

Architecture Compliance

Architecture Compliance manages scheduled or on-demand audits of CMDB data to determine which configuration items (CI) match expected attributes. The compliance audits check servers to ensure that their physical resources, such as CPU speed or memory, comply with certain standards.

The compliance process checks servers to ensure that their resources, such as CPU speed or memory, comply with standards set by your organization. Audit reports show any discrepancies in the attributes of the target CIs, and ServiceNow automatically assigns follow-on tasks to qualified users who can remediate those discrepancies.

The administrator responsible for compliance checking creates template definitions of expected attributes and then schedules an audit to check CIs for compliance. The audit results identify CIs that pass certification and itemize the discrepancies in those CIs that fail. ServiceNow automatically generates and assigns follow-on tasks to track the process of getting the CIs back into compliance. Users with the admin role activate Architecture Compliance.

Architecture Compliance roles

To access or configure certification elements, a user must have the certification_admin role. These users can create, update, and delete filters if they have the proper access to necessary tables.

In the base system, certification_admin users have limited system rights and do not have access to all the necessary tables. When assigning compliance resources, make sure to grant additional roles to the certification_admin user as needed. For example, the certification administrator needs roles that grant access to these tables:

- Company [`core_company`]
- Cost Center [`cmn_cost_center`]
- Schedule [`cmn_schedule`]

Architecture Compliance Process

Perform these tasks in this order to certify configuration items with Architecture Compliance.

1. Create a filter.

   Create a filter that defines a subset of configuration items to certify. You can create multiple versions of a filter, and then activate the version you want to use for compliance checking. Architecture compliance only supports filters on the Configuration Item [cmdb_ci] table and all tables that extend it.

2. Create a template.

   Create template conditions using values from reference fields in a related list or conditions that define the expected physical attributes of each CI in an audit. The template uses a filter to determine which configuration items the system examines based on these conditions.

3. Create and run an audit.

   Create and schedule an audit or run an audit on demand. The audit generates a set of results based on the conditions in the template you specify.

4. View audit results.

   View the audit results which display any discrepancies between the expected state, as expressed by the template conditions, and the actual state of the target configuration items.

5. Correct discrepancies.

   Correct the discrepancies the audit found by completing the follow-on tasks created by the system.

Architecture Compliance Overview module

The Architecture Compliance Overview module displays various architecture compliance reports. The Overview module is a type of homepage.

Only compliance users with certain roles can access the Overview module. The different levels of access are:

Access levels per role

Role	Access
certification	View (view overview page and refresh reports)
certification_admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) View, customize
admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) • Edit (can edit reports)

Using the Architecture Compliance Overview module

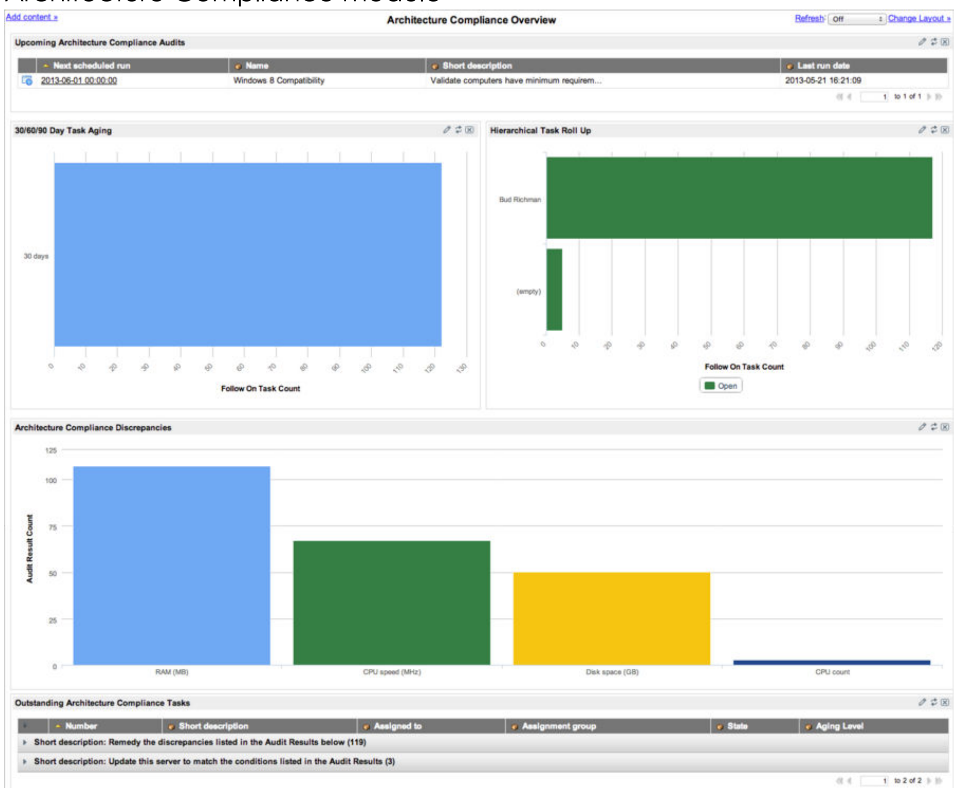
To use the Architecture Compliance Overview module, navigate to **Compliance > Architecture Compliance > Overview** and click elements within the gauges to obtain more information.

The available reports are:

Architecture Compliance Overview module gauges

Report	Description	Table
30/60/90 Day Task Aging	All outstanding follow-on tasks grouped by age in 30-day increments	Certification Task
Architecture Compliance Discrepancies	All audited attribute discrepancies	Audit Results
Hierarchical Task Roll Up	All follow-on tasks grouped by **Assigned to** user	Follow On Task
Outstanding Architecture Compliance Tasks	All follow-on tasks in the **Pending**, **Open**, or **Work in Progress** state	Follow On Task
Upcoming Architecture Compliance Audits	All scheduled audits	Audit

Architecture Compliance module



Desired State

Desired State performs scheduled or on-demand audits of CMDB data to determine which records match the expected attributes, CI relationships, and relationships to other records in the system.

For example, desired state can determine if a computer has a license for a particular software program. The compliance process checks configuration items (CI) to ensure that their attributes and relationships comply with standards set by your organization. Audit results show any discrepancies in the desired state of a record, and ServiceNow automatically assigns follow-on tasks to qualified users who can remediate those discrepancies.

Desired State roles

To access or configure certification elements, a user must have the certification_admin role. These users can create, update, and delete filters if they have the proper access to necessary tables.

In the base system, certification_admin users have limited system rights and do not have access to all the necessary tables. When assigning compliance resources, make sure to grant additional roles to the certification_admin user as needed. For example, the certification administrator requires roles that grant access to these tables:

- Company [`core_company`]
- Cost Center [`cmn_cost_center`]
- Schedule [`cmn_schedule`]

Desired State process

The desired state certification process can mean checking servers to ensure that their physical resources, such as CPU speed or memory, comply with certain standards. This process also ensures that all critical business services have a manager, support group, and approval group assigned.

The administrator responsible for certification creates definitions of desired states and then schedules an audit to check CIs for compliance. The audit results identify CIs that pass certification and itemize the discrepancies in those CIs that fail. The ServiceNow system automatically generates follow-on tasks to track the process of adjusting the CIs to the desired state.

Desired state differs substantially from data certification. Data certification is a manual process to ensure that your data matches reality. Desired state examines the same data and determines when the configuration of each item is in the desired and approved state.

1. Create a certification filter: Create a filter that defines a subset of configuration items to certify. You can create multiple versions of a filter, and then activate the version you want to use for certification. You can create filters on the Configuration Item [`cmdb_ci`] table and all tables that extend it.

2. Create a template: Create a template with conditions that define the desired state of the physical attributes, related records, and relationships for a CI. The certification filter you select for the template determines which configuration items the system examines.
3. Create and run an audit: Create an audit using the template. Set the audit to run on a schedule or on demand. The audit generates a set of results based on the conditions from the template you specify. Determine usage of follow-on tasks:
    - Determine if the audit creates follow-on tasks and assignment.
    - Determine if the same follow-on task is used for the same audit failure across multiple runs. The system property glide.allow.new.cert_follow_on_task is set to true by default, allowing for new follow on tasks to be created for the same failure, at each audit run (this system property applies only to audits that aren't scripted).
4. View audit results: View the audit results which display any discrepancies between the desired state, as specified by the template, and the actual state of the target configuration items.
5. Correct discrepancies: Correct the discrepancies the audit found by completing the follow-on tasks created by the system.

Desired State Overview module

The Desired State Overview module displays various desired state reports. The Overview module is a type of homepage.

Desired State Overview module roles

Only compliance users with certain roles can access the Overview module. The different levels of access are:

Access levels per role

Role	Access
certification	View (view overview page and refresh reports)

Role	Access
certification_admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) View, customize
admin	• View (view overview page and refresh reports) • Customize (refresh, add, delete, and rearrange reports) • Edit (can edit reports)

The different levels of access are:

- View: can view the overview page and refresh reports.
- Customize: can refresh, add, delete, and rearrange reports.
- Edit: can edit reports.

Use the Desired State Overview module

The Desired State Overview module displays various desired state reports.

Before you begin

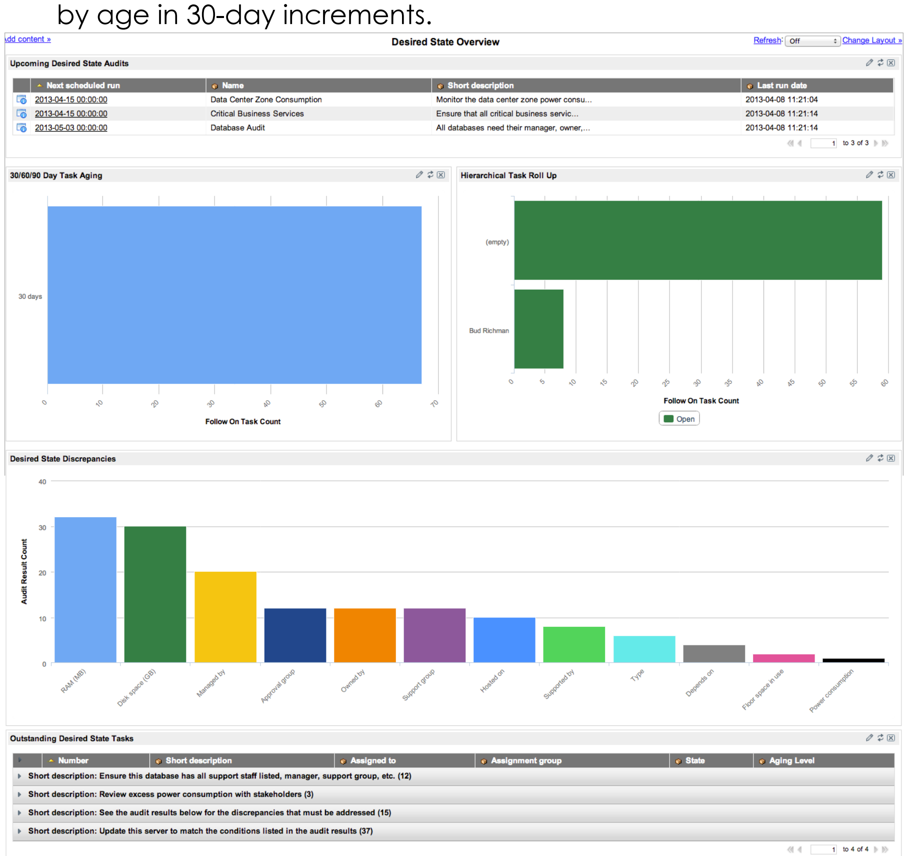
Role required: none

Procedure

1. Navigate to **All > Compliance > Desired State > Overview**.
2. Move or add reports where needed.
3. Click elements within the reports to obtain more information.

The Desired State Overview Module in the base system contains these reports:

- Upcoming Desired State Audits: All scheduled audits.
- Outstanding Desired State Tasks: All follow-on tasks in the **Pending**, **Open**, or **Work in Progress** state.
- Hierarchical Task Roll Up: All follow-on tasks grouped by **Assigned to** user.
- Desired State Discrepancies: All audit discrepancies for attributes and relationships.
- 30/60/90 Day Task Aging: All outstanding follow-on tasks grouped by age in 30-day increments.



Desired State reporting

The Desired State application includes reports to assess your audit results.

These reports are available to all users whose role gives them access to the Reporting application. Users with the admin role can share these reports with specific users of groups or change the display options.

Navigate to **Reports > View / Run**. In the Reports search field, enter all or part of the report name such as `Desired State`. You can also scroll to the designated category and select one of the reports.

In addition to these reports, you can generate your own reports.

Desired state report table

Report	Description	Category
Desired State Discrepancies	This report displays all desired state audit results that have a follow-on task that is not yet in the Closed Complete state. This report displays by column name. • Type: bar chart • Table: Audit Result [cert_audit_result]	Audit Result
Desired State Result with Stability Unstable	This report displays all audit results where the Stability field has the value Unstable. This report displays by CI and stacked by audit. • Type: bar chart	Audit Result

Report	Description	Category
	• Table: Audit Result [cert_audit_result]	
Desired State Result with Threshold Exceeded	This report displays all audit results where the Threshold field has the value Exceeded. This report displays by CI and stacks by each audit. • Type: bar chart • Table: Audit Result [cert_audit_result]	Audit Result
Upcoming Desired State Audits	This report displays the desired state audits that are scheduled to run in the next two quarters. • Type: List (tabular) report • Table: Audit [cert_audit]	Audit
30/60/90 Day Desired State Task Aging	This report displays the number of follow-on tasks that are not Closed Complete for desired state audit types. The report is grouped by aging level. • Type: Horizontal bar chart	Follow On Task

Report	Description	Category
	• Table: Follow On Task [cert_follow_on_task]	
Desired State Hierarchical Task Roll Up	This report displays similar data to the Task Aging report, but groups the results by manager. • Type: Horizontal bar chart • Table: Follow On Task [cert_follow_on_task]	Follow On Task
Outstanding Desired State Tasks	This report displays similar data to Task Aging report, but groups the results by short description. • Type: List (tabular) report • Table: Follow On Task [cert_follow_on_task]	Follow On Task

Certification audits

A certification audit compares the actual attributes of certain ServiceNow records. This audit selects a filter, against the expected attributes, relationships, and related record values defined by template conditions or a script.

You can configure the audit to create and assign follow-on tasks to remediate any discrepancies the audit finds. Audit records use a standard ServiceNow scheduler to determine when to run. After an audit runs, the results and follow-on tasks appear in related lists in the audit record.

Users with the certification_admin role can create, update, delete, and run audits. Users with the certification role can view audits, audit results, and follow-on tasks.

Create a compliance audit

Create a compliance audit. Compliance offers two types of audits: one uses templates to define conditions and the other uses a script.

Before you begin

Role required: certification_admin

Procedure

1. Ensure that an appropriate template record was created for this audit.

   Note: Conditions in the template define the values to audit.

2. Use the CI Class Manager:

   a. Navigate to **All > Configuration > CI Class Manager**.

   b. Select a class from the **Class Hierarchy**.

   c. In the sidebar, check **Advanced** and then select **Audit** in the **Compliance** group.

3. Or, navigate to one of these modules:

   - **All > Compliance > Audits**
   - **All > Compliance > Architecture Compliance > Audits**
   - **All > Compliance > Desired State > Audits**
   - **All > Compliance > Scripted Audits > Audits**

4. Select **New**.

The system opens a new record for the audit type associated with the navigation path you selected. The **Audit type** field is read-only.

5. Complete the form using the fields described in the table below.
6. Right-click the header bar and select **Save**.

   The Audit Results and the Follow On Tasks related lists appear on the form.
7. To run the audit immediately, select **Run Audit**.

   When template audits run, ServiceNow updates the date and time in the **Last run date** field and populates the related lists. For scripted audits, the **Last run date** field is not populated.
8. View the records that passed and the discrepancies found by the audit in the **Audit Results** related list.
   You can open template records and any follow-on tasks directly from this related list. Notice that the value in the **Task description** field appears as the **Short description** in the follow-on tasks.

   Note: You cannot delete audit records that have audit results or audit results that have follow-on tasks. ServiceNow disables the **Delete** option in records and lists where these dependent records exist.

Creating Audits

Field	Description
Name	Name for this audit.
Filter	Filter to use when the audit type is Scripted. This field is required for scripted audits, but is hidden for all other audit types.
Template	[Required] Template to use when this audit runs. Audit type filters the list of available templates, and only the active versions of templates are available for selection. For

Field	Description
	example, when you create an audit from Desired State, only templates of the Desired State audit type are available for selection. For the Desired State and Architecture Compliance audit types, only templates for tables that extend the Configuration Item [cmdb_ci] table are available. This field is hidden when the audit type is **Scripted**.
Table	[Read-only] Table for the template.
Create tasks	Option to create follow-on tasks for correcting discrepancies (selected). In a scripted audit, you can create the logic for either task state by using **true** to create tasks or **false** to not create tasks. By default, this check box is cleared (**false**) in a new audit record.
Assignment type	Method for assigning follow-on tasks. This field is visible only when the **Create task** check box is selected. Choices are: • User Field: Select a user reference field on the table being audited. For example, you choose the user identified in the **Managed by** field on the failed record to perform the tasks. This selection displays the **Assigned to** and **Assign to empty** fields. If the reference field on the record is empty,

Field	Description
	the value in the **Assign to empty** field is used. • Specific User: Select a specific user to perform the tasks. This selection displays the **User** field. • Group Field: Select a group reference field on the table being audited. For example, you choose the group identified in the **Support group** field on the failed record to perform the tasks. Tasks are assigned to all members of the group. This selection displays the **Assign to group** and **Assign to empty** fields. If the reference field on the record is empty, the value in the **Assign to empty** field is used. • Specific Group: Select a specific group to perform the tasks. This selection displays the **Group** field. All members of the selected group are assigned to the tasks.
User	Specific user this audit assigns to follow-on tasks. This user must have the certification role. This field is available under these conditions: • Assignment type is set to **Specific User**.

Field	Description
	• Assign to empty is set to **Create Assigned Task**, and **Assignment type** is set to **User Field**.
Assign to group	Group field that defines which group this audit assigns to the follow-on task. This field is available only when the **Assignment type** is **Group Field**.
Group	Specific group this audit assigns to follow-on tasks. This field is available only when the **Assignment type** is **Specific Group**.
Assign to	User field that defines which user this audit assigns to the follow-on task. This field is available only when the **Assignment type** is **User Field**.
Assign to empty	Behavior to use if the field selected in **Assign to** or **Assign to group** is blank on the record being audited. For example, if a follow-on task must be assigned to a manager, but no manager is identified, the **Assign to empty** setting determines what happens. This field appears only when the **Assignment type** is **User Field** or **Group Field**. Choices are: • Do Not Create Task: No follow-on task is created when the **Assign to** or **Assign to group** field is empty.

Field	Description
	• Create Unassigned Task: Create a follow-on task, but do not assign it to any user or group. The task can be manually assigned later. • Create Assigned Task: Create a follow-on task and assign it to the user or group specified. If the assignment type is **User Field**, the **User** field becomes available. If the assignment type is **Group Field**, the **Group** field becomes available. The audit automatically creates follow-on tasks for all records that have **Assign to** populated, regardless of the **Assign to empty** setting.
Short description	Brief description of the purpose of the audit.
Task description	General description of the work required for the follow-on tasks for the audit. All follow-on tasks created by this audit inherit this description.
Active	Activation control for this audit record. Clear this check box to prevent this audit from running and creating follow-on tasks.
Run	How often to run the schedule that generates the audit. • Daily • Weekly

Field	Description
	• Monthly • Periodically • Once • On Demand
Day	• If **Run** is **Weekly**, the day of the week when the audit runs. • If **Run** is **Monthly**, the day of the month when the audit runs. If the day is 29, 30 or 31, for shorter months the audit runs on the last day of the month.
Repeat Interval	If **Run** is **Periodically**, the frequency that the audit runs, based on a 24-hr. clock. Enter the number of days between audits and the time of day that you want the audit to run. For example, set **Days** to **10** and **Hours** to **14:00:00** to run the audit every 10 days at 2:00pm.
Starting	If **Run** is **Periodically** or **Once**, the date and time when the audit runs.
Time	If **Run** is **Daily**, **Weekly**, **Monthly**, or **Once**, the time of day, on a 24-hour clock, when the audit runs.
Last run date	[Read-only] The last date and time the audit ran, either on its regular schedule or manually.

Field	Description
	Audit previews do not update this field.
Next scheduled run	[Read-only] The next date and time when the audit runs. The system recalculates this field when you change the schedule.
Audit type	[Read-only] The type assigned to this audit. The system selects the audit type based on the application from which the audit is created. The type can be: • Desired State • Architecture Compliance • Compliance • Scripted
Health window	Duration of the evaluation period for threshold and stability. The health window value defines the number of Health window units in an evaluation period for an audit. This value is expressed as a positive integer. The default value for this field is **7**.
Health window unit	Unit of measurement that defines the duration of a health window. The default value for this field is **Days**. Choices are: • Minutes • Hours • Days

Field	Description
	• Months
Threshold count	Acceptable number of audit failures for the desired state field that can occur within the specified health window for a CI. The audit results indicate when a desired state field is within or has exceeded this threshold limit. The default value for the threshold is 5.
Stability count	Acceptable number of times that audit results for a CI can switch between **Certified** and **Failed** within the specified health window. The audit results for a CI indicate whether it is stable or unstable. The default value for stability is 1.
Run this script	Audit script to run which contains the conditions that a CI need to comply with to pass the audit. This field is available only when the audit type is **Scripted**. The Audit form includes a sample script with instructions for performing the audit and generating the follow-on tasks.

Copy an audit

New audits can be created from an existing audit.

1. Open the audit record you want to copy.
2. Change the name or short description to distinguish this audit from the original.
3. Make any other changes you need.

4. Right-click in the header bar and select either Insert or Insert and Stay from the context menu.
   The system clears the **Last run date** field and inserts the record into the database.

Schedule and run an audit

The system performs audits automatically from the schedule you configure.

Users with the certification_admin or admin role can generate on-demand audits directly from the Audit form by clicking **Run Audit**. When an audit runs, ServiceNow populates the **Audit Results** related list in the form and shows follow-on tasks, if any, in the **Follow On Tasks** related list. Click **Preview Audit Results** to generate an audit preview that tests your template conditions without generating any audit results.

Certification audit results

Audit results show the records that have passed or failed an audit and itemize any discrepancies detected.

A discrepancy is considered any departure from the expected conditions defined in the template or script used for the audit. Audit results provide links to the source records and to the follow-on tasks for bringing failed records into compliance. Records that pass an audit have a single entry in the results table with a state of Certified. Records that fail an audit show all discrepancies, each with a state of Failed.

ServiceNow displays results from a certification audit in these locations:

- Audit Results list
- A related list in the Audit record
- A related list in the compliance view of a CI record

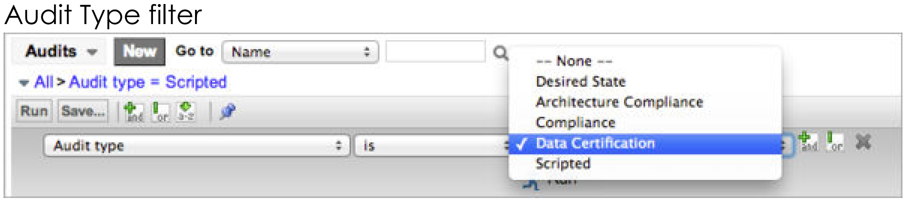
View an audit result

To generate certification results, you must first create and run an audit.

Before you begin

Role required: none

Procedure

1. Navigate to one of the following locations:
    - **Compliance > Desired State > Audit Results**
    - **Compliance > Architecture Compliance > Audit Results**
    - **Compliance > Scripted Audits > Audit Results**
    - **Data Certification > Schedules > Audit results**
2. From any audit results list, you can edit the filter to show the results for any audit type.
   Audit Type filter
   

   The results filter by audit type and grouped by audit number. Within the groups, results list by date, from oldest to newest.
3. You can open the audit record, the CI record, or the follow-on tasks from this list.

    Note: The **Audit type** field was set automatically when the audit result was created and cannot be changed. For scripted audits, the audit type is set when you create the audit record.

   Audit results show this information:

   Audit Results

Field	Description
Created	Date and time the audit ran.
Document	Record that was certified, such as a configuration item (CI).

Field	Description
State	Results of certification for each condition evaluated. The three possible states are: • Certified A certified record is one that passed all conditions. The instance generates only one audit result for a certified record. • Failed Records that are not certified have an audit result for each failed condition. The **Column name**, **Desired value**, **Discrepancy value**, and **Follow on task**are only populated for failed results. • Pending A pending state indicates that the audit is incomplete. Data certification audits use this state when a result is awaiting user input.
Column name	Audited field, relationship, or related list column that did not match the expected state.
Desired value	Attribute or relationship required for this record that was not found, from the condition in the expected state template. For data certification, this column is blank if the record has a state of Failed or Pending.
Discrepancy value	Actual value of the attribute that did not match the expected state. The follow-on task, if provided, tracks resolution of this

Field	Description
	discrepancy. In a list of results for the **Data Certification** audit type, this column is blank if the record has a state of Certified or Pending.
Follow on task	Link to the follow-on task generated for remediating a discrepancy.
Audit	Link to the audit record that produced the results.
Threshold	State of an audited, desired state field with a defined failure threshold. This threshold is the acceptable number of failures for a desired state field within a specified health window and is configured in the Audit form. Possible threshold states for the results are: • In Limit • Exceeded
Stability	Stability state of a CI. Stability state is based on the number of times the audit result for a desired state field changes from Certified to Failed within a specified health window. Possible stability states are: • Stable • Unstable

Delete an audit result

While audit results can be deleted, you cannot delete an audit result that has a follow-on task associated with it.

Before you begin

Role required: none

Procedure

1. Navigate to one of these modules:
   - **Compliance > Desired State > Audit Results**
   - **Compliance > Architecture Compliance > Audit Results**
   - **Compliance > Scripted Audits > Audit Results**

   The list groups by audit name.
2. Select the check box for a result in the list, and then select **Delete** from the **Actions on selected rows** menu at the bottom of the list.

   Note: If the result record has a follow-on task, the **Delete** option is not available. If you select multiple records, some with and some without tasks, the system only deletes those records that do not have tasks.
3. Click a date/time link to see the results for a specific CI.

   Note: The **Delete** button only appears on the form if the audit result does not have a follow-on task.
4. Click **Delete**.

View an audit result in the Compliance view

After an audit has run, you can view the results and follow-on tasks from the Compliance view in the records of every CI audited.

Before you begin

Role required: none

About this task

This view is available only for systems that use the default CI classes provided with the base ServiceNow system, such as Hardware, Software, and Computer. For information about creating views, see View Management.

Procedure

1. Navigate to **All > Configuration** and open the record of a CI that was included in a compliance audit.
2. Select the view to configure by performing the appropriate action for your list version.

Version	Action
List V2	Open the context menu and select **View > Compliance**.
List v3	Open the context menu and select **Change View**, and then click **Compliance**.

The Audit Results Compliance View appears.

Audit Results Compliance View List Descriptions

Lists	Description
Passed Audit Results	Lists audits for this CI that passed without discrepancies. The information includes the versions of the template and filter used. Records are grouped first by audit, and then by creation date and time.
Failed Audit Results	Lists all failed audits for this CI. The information includes the discrepancy data, the follow-on task, and the versions of the template and filter used. Records are grouped first by

Lists	Description
	audit, and then by creation date and time.
Follow On Tasks	Lists all follow-on tasks generated from audit discrepancies for this CI.

3. Right-click the header bar and select **View > Compliance** from the context menu.

Results preview

You can preview an audit to view potential results without saving audit results or generating follow-on tasks. For example, use this feature to test template conditions for correctness without creating thousands of result records.

In an audit record, click **Preview Audit Results** under **Related Links** to show a summary of the potential audit results appears at the top of the audit record.

Previewing does not change the **Last run date** field.

Health windows

A health window is a trailing time frame in which the ServiceNow system evaluates audit results from CIs that have desired state fields defined.

The **Health window** and **Health unit** fields define each window, and ends when an audit runs. For example, an audit runs on the fifteenth of the month with a seven-day window. It evaluates the threshold values of a desired state field from the eighth to the fifteenth. When the same audit runs the next day, the system evaluates the threshold from the ninth to the 16th, and so on. The audit counts backward seven days from the current day. ServiceNow evaluates a CI threshold value for each health window, without considering the results from the previous window. As a result, the health of a CI can fail for one audit and then pass in a subsequent audit that runs in a new window.

ServiceNow evaluates stability by recording the number of times a desired state threshold value for a CI switch between **Failed** and **Certified** within the health window. In the example shown here, a 5-minute health window was set for the desired state field on a UPS unit that measures the remaining battery time. The threshold was set at 2, which allows the field to fail two audits in the same health window.

In the initial audit, the system evaluated the threshold value for the **Seconds on battery** field within a 5-minute window. This window ran from 13:52:51 to the time of the audit at 13:57:51. The desired state field showed In Limit for that audit and the second audit conducted less than a minute later. The next two audits were conducted within five minutes of the first audit and both showed that the threshold (set at 2) was Exceeded. A subsequent audit was conducted five minutes after the audit in which the desired state field threshold was first exceeded. Since the health window had moved forward enough units, the **Seconds on battery** field was within limits again with only one failure in the 5-minute window being evaluated.

Certification filters

A certification filter creates a subset of ServiceNow records to audit, typically from configuration items (CI) of a certain type, such as all UNIX servers in a specific datacenter.

However, you can define a filter for any ServiceNow table by using any set of system-supported conditions. Audited records identified by a filter for expected attributes or relationships, depending on the audit type.

You can create multiple versions of a filter, reactivate inactive versions, and select the version you want to use in a template or a certification schedule. Only the active versions of a filter are available for selection in template records. You can use a single filter for multiple certification templates or schedules.

Certification filters

Filter	Description
Data Certification	Validates CMDB data.
Architecture Compliance	Manages reviews of CMDB data in architecture compliance audits to determine which configuration

Filter	Description
	items (CIs) match expected attributes.
Desired State	Manages reviews of CMDB data to determine which CIs match a desired state for both attributes and relationships.
Compliance	Manages reviews of records from any ServiceNow table to determine which records match an expected set of attributes and related record conditions.
IT Governance Risk and Compliance	Generates audits and tests to ensure that controls are being followed and creates tasks to track corrective actions.

Roles

In the base ServiceNow system, users with the certification_admin role have limited system rights and do not have access to the tables required for creating a filter.

When assigning compliance resources, make sure certification_admin users have any additional roles they need. For example, a user requires roles that grant access to the Company [`core_company`] table.

Compliance filter

The compliance filter for license bases uses the following fields to define entitled users or CIs.

These field values can be used independently or together to calculate compliance.

Compliance Filter

Value	Description
Entitled company	All users or configuration items (CI) at all locations of this company, in all departments are entitled to use this software package. Compliance at this level calculates how many licenses are purchased for the company at large and how many entitled users or CIs consume them.
Entitled location	CIs and users who are assigned to this company location in any department are entitled to use this software package. Compliance at this level calculates how many licenses are purchased for this company location and how many users and CIs consume them.
Entitled department	Only the users or CIs in this department at this company location are entitled to use this software package. Compliance is calculated for a single department only.

The license form can display information about all CIs or named users who are using this software package. The form indicates when license reconciliation is necessary and displays all compliant users or CIs.

Possible compliance levels are:

Compliance

Level	Description
Non applicable	Compliance levels for all infrastructure licenses that are related to cluster licenses are set

Level	Description
	to **Non applicable** automatically. Compliance levels are calculated in the cluster license only, and not in the related infrastructure licenses.
Out of compliance	More licenses are being consumed than were purchased. There are more users or CIs using this license than the license allows, and some users or CIs are not be entitled to use this software package.
Unused	The licenses for this software package are currently unused.
Reconciliation required	CIs or users who are not entitled to use this software are consuming licenses. Licenses that require reconciliation are considered out of compliance. Reconciliation requires action to ensure that unentitled users are not using the software. Reconciliation involves uninstalling software or increasing license counts to match actual user counts.
Nearly out of compliance	For a software package to be at this compliance level, more than 95% of the licenses are in use by entitled users or CIs. License bases at this level are considered to be **In compliance**.
In compliance	This software package has unused licenses. All users or CIs using a license are entitled to use this software package.

Create a filter

You can create as many versions of a filter as necessary. You can then designate which versions are active and available for selection in Compliance template records, Governance Risk and Compliance control test definitions, or Data Certification schedule definitions.

Before you begin

Role required: none

Procedure

1. Navigate to **All > Configuration > CI Class Manager**, and:
   a. Click **Hierarchy** to display the CI Classes list.
      Select the class to create a filter for.
   b. In the class navigation bar, expand **Health**, select **Compliance**, and then click **Certification Filter**.
2. Or, navigate to one of these modules:
   - **Compliance > Filters**
   - **Data Certification > Schedules > Certification Filters**
3. Select an existing filter to edit, or click **New**.
4. Fill in the fields (see table below).
5. Click **Submit**.
   This action saves the filter as version 1.
6. To create another version of this filter, open the record and modify the name, table, or conditions.

   Note: You can change a filter Description without incrementing a version.

7. Click **Update**.

   The system saves a new version of the current filter and makes it the Active version. The previous version is marked inactive. The system displays only active filter versions for selection when you create templates or schedules.

Creating Filters

Field	Description
Number	[Read-only] Displays the automatically assigned filter identification number. All versions of a filter have the same number.
Name	[Required] Filter name.
Description	[Optional] Describes this filter. You can change the description of a filter without incrementing a version.
Table	Specifies the table containing the records to select. The template or schedule that uses this filter works on this table. For example, select the ESXi Server [`cmdb_ci_esx_server`] table to select VMware ESX servers.
Active	Makes this filter available for use from the **Filter** field on the Certification Template or Schedule Definition form. Multiple versions of a filter can be active. You can activate or deactivate a filter without incrementing the version.
Version	[Read-only] Indicates the version of this filter. Any changes to this filter, except to the description or the **Active** check box, makes it inactive. The system increments the version of the updated filter

Field	Description
	and marks it as active. The system saves all versions of the filter and makes them available for reactivation.
Filter condition	Specifies the fields, operators, and values that create the filter. The available fields are based on the selected table. The condition builder shows the number of records that match the conditions. Click the refresh icon Refresh Conditions to recalculate the number of matching records when you edit the conditions.

Copy a filter

New filters can be created from an existing filter.

Before you begin

Role required: none

Procedure

1. Navigate to **All > Configuration > CI Class Manager**, and:
    a. Click **Hierarchy** to display the CI Classes list. Select the class to create a filter for.
    b. In the class navigation bar, expand **Health** and select **Compliance**. Then click **Certification Filter**.
2. Or, navigate to one of these modules:
    - **Compliance > Filters**

- **Data Certification > Schedules > Certification Filters**

3. Open the filter record that you want to copy.
4. Make sure to change the filter name or description to distinguish the new filter from the original.
5. Make any other necessary changes.
6. Right-click in the header bar and select either **Insert** or **Insert and Stay** from the context menu.
   The system increments the record number and sets the version to 1 for the new record. Both the original filter and the copy are Active and appear in the record list. Showing all copies of a filter allows you to see the entire history of the filter.

Delete a certification filter

Delete certification filters that are no longer needed and no longer in use.

Before you begin

Role required: certification_admin or admin

About this task

You can't delete a certification filter that is being used in a template or a scripted audit.

Procedure

1. Navigate to **All > Configuration > CI Class Manager**, and:
   a. Select **Hierarchy** to show the CI Classes list.
   b. Select the class to delete a filter for.
   c. In the class navigation bar, expand **Health**.
   d. Select **Compliance** and then **Certification Filter**.
2. Or, navigate to one of these modules:

- **Compliance > Filters**
- **Data Certification > Schedules > Certification Filters**

3. Open the filter record you want to delete.
    a. To delete a single filter version, open that version record and click **Delete**.

       The system hides the **Delete** button for filters that are in use. If you delete the latest version of a filter that is active, the previous version of that filter is reset to Active.
    b. To delete all unused and inactive versions of a filter, open any version of that filter and click **Delete inactive versions** under **Related Links**.
4. When prompted, select **OK** to proceed.
   The system deletes unused filter versions. A message in the header bar identifies filter versions that cannot be deleted because they are used in a template or scripted audit.

Manage a filter version in a form

You can view and manage all versions of a filter from the Certification Filter form.

Before you begin

Role required: none

About this task

Versions can be displayed in a list. The default list of filters displays only the active version of each filter. To see all filter versions in the list view, select **All** in the breadcrumbs.

Procedure

1. Open any version of a filter.
   The Other Versions related list displays all other versions of this filter, both active and inactive. The system prevents you from editing either the filter version or the record number in the list view.

2. Click any version in the related list to display the record for that version.
3. To make an inactive filter the current version, open the filter, edit it if desired, and then click **Revert**.
   This action:
   - Deactivates the previous active version of the filter.
   - Copies the inactive filter.
   - Makes this new copy current and active.

Certification follow-on tasks

The ServiceNow system can automatically generate and assign follow-on tasks to correct discrepancies detected during compliance audits.

The system property glide.allow.new.cert_follow_on_task is set to true by default, allowing for new follow on tasks to be created for the same failure, at each audit run. You can set this property to false, to configure audit to use the same follow-on task for the same audit failure across multiple runs.

The system property glide.allow.new.cert_follow_on_task applies only to audits which aren't scripted.

The system property glide.allow.new.cert_follow_on_task applies only to audits which aren't scripted.

You configure and assign follow-on tasks to qualified users or groups in the audit record. A user with the certification_admin role can reassign any follow-on task. The Audit Results related list in the Follow On Task form contains links to the records that failed.

Access follow-on tasks

Users with the certification role can only access follow-on tasks assigned to them but can reassign these tasks to other users.

1. Navigate to **All > Compliance > My Follow On Tasks**.
   The list contains all active follow-on tasks assigned to the logged in user.

2. Open a task.

   The record shows the specifics of the task, the task activity, and the failed audit results.

3. Open records from the **Audit Results** related list to see each discrepancy.

4. Go to the CI named in the record and perform the work to bring it into compliance.

5. Update the **State** field in the follow-on task record and add work notes as you correct each discrepancy.

   When you change the state, the system updates the task activity appropriately.

   When the task is **Closed Complete** it no longer appears on the **My Work** list.

Manage follow-on tasks

Users with the certification_admin or admin role can see all follow-on tasks.

Before you begin

Role required: none

About this task

Tasks are pre-assigned to a user or group as specified in the audit record, but users with the certification_admin role can reassign the task.

Procedure

1. Navigate to the appropriate application:

   - **Compliance > Architecture Compliance > Follow On Tasks**
   - **Compliance > Desired State > Follow On Tasks**
   - **Compliance > Scripted Audits > Follow On Tasks**

The list of follow-on tasks appears, filtered by audit type.

2. Open a task.

   The **Audit** and **Configuration item** fields are read-only for all users.

3. Edit the **Change group** or the **Assigned to** field if necessary.
4. Edit the **Short description** field if necessary.

   The short description is inherited from the **Task description** field in the Audit form.

5. Use the links in the **Audit Results** related list to open the individual records that failed the audit.
6. If you update the follow-on task record, be sure to add work notes.

Certification templates

Certification templates can define attributes, relationships, and reference field values that indicate what a record is expected to contain.

These values are used to perform audits on ServiceNow records. The certification filter selected in the template identifies the table and records to audit, and the template conditions set the expected state for those records. The type of audit you create determines which tables and template conditions are available.

Users with the certification_admin role can create, update, and delete templates. Users with the certification role can view template versions.

Certification template audit types

When you create a template, ServiceNow assigns an Audit type that determines which tables and conditions are available in the certification template. This value is based on the application from which the template is created. Each application lists only the templates with the associated type.

Available Condition Builders

The available condition builders for each audit type:

- Compliance: Runs audits on any set of ServiceNow records, not only configuration items (CI). This audit type provides the following types of conditions for any ServiceNow table:
  - Attribute: Sets conditions for the attributes of the records.
  - Related List: Runs audits on records in tables that reference the table defined in the template.
- Architecture Compliance: Defines the following types of conditions for tables that extend the Configuration Item [cmdb_ci] table.
  - Attribute: Sets conditions for physical attributes of CIs, such as memory or disk size.
  - Related List: Runs audits on records in tables that reference the table defined in the template.
- Desired State: Defines the following types of conditions for tables that extend the Configuration Item [cmdb_ci] table.
  - Attribute: Sets conditions for physical attributes of CIs, such as memory or disk size.
  - CI relationship: Defines the relationships these CIs have with other CIs. An example of a relationship is a business service, such as Outlook Web Access, that depends on a server.
  - User relationship: Defines the user who reviewed the log records. The only operator available with this condition builder.
  - Group relationship: Defines user groups who backed up this CI. The only operator available with this condition builder.
  - Related List: Runs audits on records in tables that point toward the table defined in the template.

Certification Template Record List

The default Templates list displays only the active version of each template, but you can update the breadcrumbs to display all template versions.

- Default Templates List: The default Templates list displays only the active version of each template, filtered by **Audit type**.

- All Template Versions: To view all template versions for an audit type, click the arrow before **Active=true** to remove that condition from the breadcrumbs. 

Create or edit a certification template

To create a certification template, follow these instructions.

Before you begin

Activate the Certification Core plugin to enable the Compliance functionality. See Compliance Activation for details.

Procedure

1. Ensure that you have an appropriate filter that defines the records the template evaluates.

   The template applies its conditions to these records.

2. Use the CI Class Manager to navigate to the Certification Template form:

   a. Navigate to **All > Configuration > CI Class Manager**.

   b. Select **Hierarchy** to display the CI Classes list.
      Select the class for which to create a certification template.

   c. In the class navigation bar, expand **Health** and then select **Compliance**.

3. Select **Certification Template**.

4. Or, navigate using one of these paths:

   - **All > Compliance > Architecture Compliance > Templates**
   - **All > Compliance > Desired State > Templates**
   - **All > Compliance > Templates**
   - **All > Audit Definitions > Templates**

5. Select **New** or select a certification template to edit.

The following fields are completed automatically:

- **Number**: Each new template has a unique number. All versions of the same template use the same number.
- **Active**: All new templates are set to Active.
- **Version**: The version of a new template is set to 1.
- **Audit type**: The system sets the default type to Architecture Compliance, Desired State, or Compliance, depending on the application in which the template was created. You can select a different type when you create the template, but the field becomes read-only when you submit the record. The system uses audit types to filter record lists for appropriate data and determine which conditions are visible on the template form.

6. Complete the following mandatory fields.
    - **Name**: Enter a descriptive name for this template. The name helps identify the purpose.
    - **Filter**: Select the filter that identifies the records to be certified. You can select either active or inactive filter versions. By default, the system presents only active versions for selection. If you start enteringthe name of a filter, the auto-complete feature displays all versions for selection. For architecture compliance and desired state templates, only filters that use a table extended from Configuration Item [cmdb_ci] appear on the choice list. All filters appear on the choice list for a compliance template. After you select a filter, the template condition builder appears. The template operates on the table specified in the filter.
7. Enter a description for this template.
8. Define certification conditions using the condition builders.

    All conditions are AND conditions. The audit type of the template determines which conditions are available.

    - **Certification Attribute Conditions**: [All audit types] Select configuration item attributes or specifications to certify, such as CPU count, memory, or disk space. Available fields in the attribute condition builder depend on the table from the filter. Typical ServiceNow conditions for attributes are available, including the between operator for setting numerical conditions with high and low boundary values. This operator was added specifically for desired state conditions.

The **Show Related Fields** item supports dot-walking, enabling you to include referenced fields in a certification attribute condition. Select **Show Related Fields** or **Remove Related Fields** to add or remove referenced fields (in the form of <field> => <field>). Select a referenced field to drill down to the next level of referenced fields.

See Dot walking.

- **Certification CI Relationship Conditions**: [Desired State audit types] Define the CI relationships to certify, such as Runs on or Depends on.
- **Certification User Relationship Conditions**: [Desired State audit types] Select the desired user relationship for this configuration item. The relationship provided in the base system is Log reviewed by.
- **Certification Group Relationship Conditions**: [Desired State audit types] Select the desired group relationship for this configuration item. The relationship provided in the base system is Backed up by.
- **Certification Related List Conditions**: [All audit types] Select field values from tables that reference the template table, or user-defined related lists ,which are created via custom relationships in the sys_relationship table. To create a condition that evaluates all servers in the Server [cmdb_ci_server] table for the presence of Microsoft Word 2007, as referenced in the Software Installation [cmdb_sam_sw_install] table. The resulting condition is [Software Installation->Installed on] [Display name] [is] [Microsoft Word 2007].

  Check **All** to include all records in the condition requirements of the related list. If there are no records in the related list, then:

  - If **All** is checked, the condition requirement is met.

- If **All** is cleared, the requirement is not met.

Note: By default, the condition builders for relationships display only suggested relationships. To see all possible relationships, select the **Show all relationships** check box on the side of the form.

a. Select **Insert a new row** to insert a condition.
You cannot insert an empty condition.

b. Select the green check mark icon to save a condition.
Make sure to save the condition before performing any other operation. Updating the form does not save the condition.

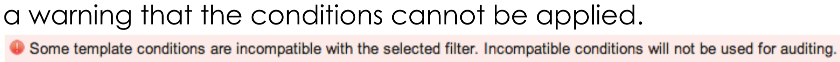
c. To delete a condition, select the red **X** beside the condition.
The system marks the condition as inactive.

d. To reactivate a condition, select the gray **X**.
If another condition for the same field exists, the system prevents reactivation and warns you of the conflict.

9. Select **Submit**.
ServiceNow saves the template as version 1.

10. To create another version of the template, change the name, edit the conditions, or select a different filter.
Updating the template description does not create a new version.

Note: If you select a filter whose table is incompatible with the existing template conditions, the system displays a warning that the conditions cannot be applied.



11. Select **Update**.
The system saves a new version of the current template and makes it the Active version. The previous version is marked inactive.

Clone a Certification template

New templates can be cloned from an existing template.

1. Open the template record to be copied.

2. Make any necessary changes.

3. Change the template name or description to distinguish it from the original.
4. Click **Clone**.
   ServiceNow increments the record number above the highest template number and sets the version of the new record to 1. A message appears under the header bar naming the source record for the clone.Both templates are Active and appear in the record list. The record list allows you to see the entire history of the template.

Manage Certification template versions

You can view and manage all versions of a template from the Template form.

1. Open any version of a template.
   The **Other Versions** related list displays all other versions of this template, both active and inactive.
2. Click any version in the related list to display the record for that version.
3. Update the template to create a new version.
   The system increments a version of the template when you edit any field except **Description** and **Active**. You can manage the template versions without returning to the list view.
4. To make an inactive template the current version, open that version, edit it if desired, and then click **Revert**.
   This action does:
   - Deactivates the previously active version of the template.
   - Copies the inactive template.
   - Makes the new copy the current, active version.
5. Select the **Audits** related list to view all audits configured to use this template.
6. Click **New** to create a new audit record with the template selection and table pre-populated.

Delete a Certification template

Certification templates can be deleted.

About this task

Only users with the certification_admin or admin role can delete template versions. You cannot delete a template version that is being used for an audit.

Procedure

1. To delete a single template version, open that version record and click **Delete**.

   The system hides the **Delete** button for templates that are in use. If you delete the latest, active version of a template, the previous version of that template is reset to Active.

2. To delete all unused and inactive versions of a template, open any version of that template and click **Delete inactive versions** under **Related Links**.
   This control appears on all versions, whether they are used in an audit.

3. When prompted, click **OK** to proceed.
   The system deletes only template versions that are not used in an audit. All protected versions are named in a message that appears in the header bar.

Scripted audits

A scripted audit enables users with the certification_admin role to conduct an audit from a script rather than using restrictive template conditions.

A scripted audit uses a certification filter to select the records to audit, and then creates standard follow-on tasks for remediation of any discrepancies. Use this type of audit to query for any values or states that a script can define. A scripted audit is a specific audit type that is activated together with the Desired State plugin. ServiceNow provides a sample audit script with configuration instructions.

Create a scripted audit

A scripted audit is an audit whose conditions are defined by a script.

1. Navigate to **All > Compliance > Scripted Audits > Audits**.

   An audit type of Scripted filters the list.

2. Click **New**.

3. Complete the form (see table).

4. Create the audit script.

   The Run this script field includes a sample script with instructions for performing the audit and generating the follow-on tasks. This field appears only when you access audits from the Scripted Audits module.

5. Click **Submit**.

   Sample script:

```
/*
/////////////////////////////////////////////////////
/// This script works with Data Center Zones filter //
/////////////////////////////////////////////////////

var desiredFloorSpaceUsage = 30;               // Valu
e to audit against
var assignToUser = '46d44a23a9fe19810012d100cca80666';
 // Beth Anglin
var assignToGroup = '8a5055c9c61122780043563ef53438e3'
;       // Hardware group
var taskMsg = 'See the audit results below for the dis
crepancies that must be addressed';

// API call to retrieve records based on the filter
var gr = new SNC.CertificationProcessing().getFilterRec
ords(current.filter);

// Loop over all records defined by the filter
while(gr.next()) {
        var sysId = gr.getValue('sys_id');     // Sys
```

```
ID of audited record
        var floorSpaceInUse = gr.getValue('floor_space_i
n_use');        // Value to audit

        // Determine if certification condition passes
or fails
        if (floorSpaceInUse < desiredFloorSpaceUsage) {
                var columnNameSpace = gr.floor_space_in
_use.getLabel();        // String value of column audi
ted against

                // Call create Follow on Task API and
save the returned sys_id for use in logging audit resu
lt fail
                // Params:
                // auditId - Sys id of the audit recor
d executed
                // ciId Sys - id of the configuration i
tem.  Empty string if not a cmdb ci
                // assignedTo - Sys id of user to assi
gn task to.  Can be empty
                // assignmentGroup - Sys id of group t
o assign task to.  Can be empty
                // shortDescr - Short description for
the Follow On Task.  Can be empty
                // Return value: Sys id of the created
 follow on task
                var followOnTask = new SNC.Certificatio
nProcessing().createFollowOnTask(current.sys_id, sysId
, assignToUser, '', taskMsg);

                // Call log failed result API
                // Params:
                // auditId - Sys id of audit record ex
ecuted
                // auditedRecordId - Sys id of the rec
ord audited
                // followOnTask - Sys id of the follow
 on task associated with the audited record(@see audit
edRecordId). Can be empty
                // columnDisplayName - Label of the co
lumn audited(ex. Disk space (GB)).  Can be empty
                // operatorLabel - Label of the operat
or used to audit the column(ex. is not empty, greater
```

```
than). Can be empty
                // desiredValue - Desired value of the
 column.  Can be empty
                // discrepancyValue - Discrepancy valu
e.  Can be empty
                // isCI - True, if audited record is a
 CI. False, otherwise.
                // domainToUse - Sys domain of the "ce
rt_audit" record.  Can be empty
                new SNC.CertificationProcessing().logAu
ditResultFail(current.sys_id, sysId, followOnTask, col
umnNameSpace, 'greater than', desiredFloorSpaceUsage,
floorSpaceInUse, true);
        } else { // If certification condition pass, wr
ite a Audit Result Pass via API
                // Params:
                // auditId - Sys id of audit record ex
ecuted
                // auditedRecordId - Sys id of the rec
ord audited
                // isCI - True, if audited record is a
 CI. False, otherwise.  Can be empty.
                // domainToUse - Sys domain of the "ce
rt_audit" record.  Can be empty.
                new SNC.CertificationProcessing().logAu
ditResultPass(current.sys_id, sysId, true);
        }
}
*/
```

New scripted audit table.

Field	Description
Name	Name for this audit.
Filter	Filter to use when the audit type is Scripted. This field is required for scripted audits, but is hidden for all other audit types.
Template	[Required] Template to use when this audit runs. Audit type filters the list of available

Field	Description
	templates and only the active versions of a template are available for selection. This field is hidden when the audit type is Scripted.
Table	[Read-only] Displays the table for the template.
Create tasks	Creates follow-on tasks for correcting discrepancies when selected. In a scripted audit, you can create the logic for either task state by using true to create a task or false if no task is created. By default, this check box is cleared (false) in a new audit record.
Assignment type	A choice list to select how the audit assigns the follow-on tasks. This field is visible only when the Create task check box is selected. Choices are: • User Field: elect a user reference field on the table being audited. As an example, select the user named in the Managed by field on the failed record to perform the tasks. This selection displays the Assigned to and Assign to empty fields. If the reference field on the record is empty, the value in the Assign to empty field is used. • Specific User: Select a specific user to perform the

Field	Description
	tasks. This selection displays the User field. • Group Field: Select a group reference field on the table being audited. As an example, select the Support group from the failed record to perform the tasks. This selection displays the Assign to group and Assign to empty fields. All members of the group from the reference field on the failed record are assigned to the tasks. If the reference field on the record is empty, the value in the Assign to empty field is used. • Specific Group: Select a specific group to perform the tasks. This selection displays the Group field. All members of the selected group are assigned to the tasks.
User	The specific user this audit assigns to follow-on tasks. This field is available under these conditions: • Assignment type is set to Specific User. • Assign to empty is set to Create Assigned Task, and Assignment type is set to User Field.

Field	Description
	Note: Ensure that the specified user has the certification role.
Assign to group	The group field that defines which group this audit assigns to the follow-on task. This field is available only when the Assignment type is Group Field.
Group	The specific group this audit assign to follow-on tasks. This field is available only when the Assignment type is Specific Group and you have selected Group Field as the assignment type.
Assign to	The user field that defines which user this audit assigns to the follow-on task. This field is available only when the Assignment type is User Field.
Assign to empty	The behavior to use if the field selected in Assign to or Assign to group is blank on the record being audited. For example, if a follow-on task must be assigned to a manager, but no manager is identified, the value in this field determines what happens. This field appears only when the Assignment type is User Field or Group Field. The possible selections are: • Do Not Create Task: No follow-on task is created

Field	Description
	when the Assign to or Assign to group field is empty. • Create Unassigned Task: Create a follow-on task, but do not assign it to any user or group. The task can be manually assigned later. • Create Assigned Task: Create a follow-on task and assign it to the user or group specified. If you selected an assignment type of User Field, the User field becomes available. If you selected the Group Field type, the Group field becomes available. The audit automatically creates follow-on tasks for all records that have Assign to populated, regardless of which selection you make for Assign to empty.
Short description	Brief description of the purpose of the audit.
Task description	General description of the work required for the follow-on tasks created by this audit. All follow-on tasks created by this audit inherit this description.
Active	Activates this audit schedule and generates follow-on tasks at the scheduled date and time. Clear this check box to hide scheduling fields on the form

Field	Description
	(except Last run date) and not generate follow-on tasks.
Run	How often to run the schedule that generates the audit. • Daily • Weekly • Monthly • Periodically • Once • On demand
Day	• If Run is Weekly, the day of the week when the audit runs. • If Run is Monthly, the day of the month when the audit runs. If the day is 29, 30 or 31, for shorter months the audit runs on the last day of the month.
Repeat Interval	If Run is Periodically, the frequency that the audit runs entered in time, days, or both. For example, set Days to 10 and Hours to 14:00:00 to run the audit every 10 days at 2:00pm.
Starting	If Run is Periodically or Once, the date and time when the audit runs.
Time	If Run is Daily, Weekly, Monthly, or Once, the time of day, on

Field	Description
	a 24-hour clock, when the audit runs.
Last run date	[Read-only] The last date and time the audit ran, either on its regular schedule or manually. Audit previews do not update this field.
Next scheduled run	[Read-only] The next date and time on which the audit runs. The system recalculates this field when you change the schedule.
Audit type	[Read-only] The type assigned to this audit. The system selects the audit type based on the application from which the audit was created and can be: • Desired State • Architecture Compliance • Compliance • Scripted
Run this script	Audit script to run. This field is available only when the audit type is Scripted. The Audit form includes a sample script with instructions for performing the audit and generating the follow-on tasks. See Script Methods for a list of the methods provided and the accepted parameters.

Script methods

ServiceNow provides four methods for creating the audit script.

Script methods

Name	Description	Parameters
getFilterRecords	public GlideRecord getFilterRecords(String filterId)	filterID: The sys_id of the filter to use.
logAuditResultPass	public void logAuditResultPass(String auditId, String auditedRecordId, boolean isCI, String domainToUse)	auditId: Sys_id of audit record executed auditedRecordId: Sys_id of the record audited. isCI: True, if the audited record is a CI, false if otherwise. domainToUse: Sys_domain of the cert_audit record.
logAuditResultFail	public void logAuditResultFail(String auditId, String auditedRecordId, String followOnTask, String columnDisplayName, String operatorLabel, String desiredValue, String discrepancyValue, boolean isCI, String domainToUse)	auditId: Sys_id of audit record executed. auditedRecordId: Sys_id of the record audited. followOnTask: Sys_id of the follow-on task associated with the audited record and can be an empty string. columnDisplayName: Label of the column

Name	Description	Parameters
		audited. For example, Disk space (GB). operatorLabel: Label of the operator used to audit the column. For example, is not empty or greater than can be the label. desiredValue: Desired value of the column. discrepancyValue: Discrepancy value. isCI: True, if the audited record is a CI, false if otherwise. domainToUse: Sys_domain of the cert_audit record.
createFollowOnTask()	public String createFollowOnTask(String auditId, String ciId, String assignedTo, String assignmentGroup, String shortDescr)	auditId: Sys_id of the audit record executed. ciId: Sys_id of the configuration item. This string is empty when the table is not extended from the cmdb_ci table. assignedTo: Sys_id of the assigned user of the task. This string can be empty.

Name	Description	Parameters
		assignmentGroup: Sys_id of the group the task is assigned to. This string can be empty. shortDescr: The text to use for the short description of the follow-on task.